

Scaling Delivery Systems for Carp, Tilapia, and Small Indigenous Species (SIS) Genetics and Input Services in Bangladesh

Section 0 – Title and Contact Information

Title of Solution Track:

Scaling Delivery Systems for Carp, Tilapia, and Small Indigenous Species (SIS) Genetics and Input Services in Bangladesh

Flagship Alignment:

Grand Challenge 2: Deliver Inputs and Services at Scale through Inclusive Market Systems & Flagship 2.3 – Delivering Sustainable Animal and Aquaculture Genetics, Inputs, and Services

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WorldFish

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Key CGIAR Center Partners:

- International Food Policy Research Institute (IFPRI)

External Partners:

- Department of Fisheries (DoF), Government of Bangladesh
- Bangladesh Fisheries Research Institute (BFRI)
- Palli Karma-Sahayak Foundation (PKSF) (MoU process underway)
- Wildlife Conservation Society (WCS)
- Private sector seed and input enterprises
- ARITS Limited, Digital Partner
- Non-governmental organizations (NGOs)
- Food Processor (Bengal Fish, Apon Wellbing; MoA discussion underway)
- Feed Manufacturer (Spectra Hexa Feeds Ltd; MoA discussion underway)

Technical Link to Other Science Programs:

- Sustainable Animal and Aquatic Foods (SAAF)
- Better Diets and Nutrition (BDN)
- Policy Innovations Science (PIP)

Section 1 – Innovation Package, Value Proposition & Alignment

1.1 The Problem to be Solved

Aquaculture in Bangladesh underpins rural livelihoods, employment, and nutrition, contributing 59% of national fish production and supporting the diets and incomes of over 18 million people (DoF, 2024; 3ie, 2025). Yet its growth is constrained by a fragmented and low-trust seed and input delivery system. Farmers often receive degraded seed of farmed fish such as carp, tilapia, and SIS due to weak broodstock management, limited certification, and dispersed nursery networks. These weaknesses lower survival rates, slow growth, and increase disease risks, reducing farm profitability and threatening national food and nutrition security.

Although improved strains have been developed through CGIAR research, these advances rarely reach farming households. Public extension services are overstretched, and private actors remain fragmented, leaving smallholders particularly women, youth, and marginalized groups in climate-vulnerable areas without reliable access to quality seed, inputs, and advisory services.

Overcoming this constraint requires decentralized, quality-assured delivery systems that connect hatcheries, nurseries, seed and input suppliers into inclusive scaling pathways capable of reliably reaching farmers across Bangladesh.

1.2 Scaling Ambition

This Solution Track will transform aquaculture delivery in Bangladesh by embedding decentralized, quality-assured systems that expand reliable access to improved carp, tilapia, and SIS genetics and input services. **By 2026–2028**, it will enable **546,000 farmers** ($\geq 30\%$ women and youth) to adopt verified fish seed, inputs, and advisory services; **create or enhance 21,742 full-time equivalent jobs**; and generate **88,976 consumer-equivalent years of access to affordable nutritious fish**. Around **240,000 ha will be managed under good aquaculture practices**, supporting environmental sustainability. These outcomes represent the strongest contributions to Scaling for Impact KPIs on adoption of innovations, job creation, and consumer access to healthier diets.

By 2030, the ambition is to reach **1 million farmers** and further expand good aquaculture practices nationwide. Evidence from this track will guide the DoF in institutionalizing decentralized delivery within national aquaculture extension and subsidy systems, ensuring long-term sustainability.

1.3 Innovation & Value Proposition

The innovation of this Solution Track lies in scaling a decentralized delivery system, with SeedCompacts at its core, to make improved carp, tilapia and SIS genetics developed under SAAF reliably accessible to farmers. SeedCompacts provide a tested mechanism for organizing hatcheries, nurseries, and seed traders into accountable networks, supported by certification and designed to introduce digital traceability that will strengthen quality assurance and farmer trust. This structure will be reinforced by the OneFish app (developed with ARITS) for farm-specific

advisory and market linkages; mobile SIS hatcheries to extend access in coastal and remote areas; and nutrition-sensitive outreach delivered by trained community health workers (CHWs).

The novelty lies in integrating these delivery mechanisms—seed networks, digital tools, and community nutrition into a partner-led, last-mile system positioned for government institutionalization.

Value proposition:

- Institutions: stronger certification and monitoring.
- Enterprises: predictable demand and revenue.
- Farmers: reliable inputs and higher productivity.
- Women/youth: greater access and service roles.
- Consumers: affordable fish and healthier diets.

These delivery innovations are reinforced by linkages to CGIAR science programs, including SAAF (genetics), BDN (nutrition outcomes), and PIP (institutionalization with IFPRI).

1.4 Customers & Alignment

Institutional customers: DoF; PKSf and partner microfinance institutions (MFIs); donor and lender programs supporting aquaculture delivery; and private agribusiness and digital partners, including hatchery, nursery, seed and input networks, and the ARITS-developed OneFish platform, feed manufacturers like Spectra Hexa Feeds Ltd, and processors such as Bengal Fish, and market actors such as Apon Wellbeing.

End users: Smallholder pond and rice–fish farmers; hatchery, nursery, seed, and input small and medium-sized enterprises; and producer groups or cooperatives, with emphasis on women and youth in climate-vulnerable and other priority aquaculture areas.

Alignment: This track supports Flagship 2.3 under AoW2 by operationalizing partner-led delivery systems that move proven genetics and inputs to the last mile, directly contributing to the Grand Challenge on inclusive market systems through adoption, jobs, and consumer access.

1.5 Evidence of Impact

Effectiveness (impact). On-farm trials in Bangladesh show improved rohu delivering 33–37% higher harvest weights with better survival than local strains, raising farmer profitability (Hamilton et al., 2022; Ali et al., 2025). In AMD pilots (2023–24), 101 nurseries and 30 input dealers supplied improved carp and SIS seed and inputs to 10,234 farmers, covering 1,907 ha and generating approximately 610 t of additional rohu production valued at USD 1.27 million (Halder and Biswas, 2024).

NGO CHWs reached >52,000 households with fish-specific messages, improving women’s dietary diversity (Halder and Biswas, 2024).

Decentralized extension delivered by local service providers (LSPs) increased productivity (up to +25%), adoption of best practices, sales, and revenues, while enhancing household food security (Dompereh et al., 2024; Obi et al., 2025).

Scalability (uptake beyond pilots). Corporate Social Responsibility (CSR) financing (Standard Chartered Bank–CNRS–WorldFish) mobilized US\$170k to expand G3 rohu distribution through hatchery, nursery and input networks; the same structure underpins SeedCompacts within the broader decentralized model. Ongoing engagement by DoF, BFRI, and private hatcheries/nurseries in Scaling for Impact activities in 2025 demonstrates partner-led uptake beyond pilots, providing a credible pathway for policy alignment and institutionalization.

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

2.1 Scaling Pathway & Sustainability

Scaling pathways:

- **Public sector pathway.** BFRI plays a key role in co-developing standards for broodstock, performance, and biosecurity in collaboration with WorldFish and other stakeholders, while DoF ensures enforcement through certification, inspection, extension and compliance monitoring. The WorldFish Genetic Improvement Program (GIP), under SAAF, supplies improved carp, tilapia, and SIS broodstock to DoF, BFRI, and private hatcheries. S4I will link SeedCompacts with government hatcheries, enabling broodstock managed by DoF and BFRI to flow through certified nursery and trader networks, delivering trusted seed to the last mile. MELIA evidence will inform policy briefs, embedding quality assurance (QA) standards and decentralized delivery in national fisheries policy.
- **Private sector pathway.** SeedCompacts organize hatcheries, nurseries, and traders into accountable seed networks supplying verified seed and advisory services. Input dealers and retailers, while outside the compacts, extend reach by linking farmers to inputs and services. Private digital partners support the OneFish app, providing farm-specific messages, traceability, and market linkages.
- **Community pathway.** NGO health workers deliver nutrition messages to households, particularly women and mothers, linking adoption to dietary outcomes. Mobile SIS hatcheries expand access in coastal and remote areas. Bilateral initiatives, such as AABS and Sustho Sagor initiative (2025–2029, led by WCS with WorldFish), strengthen outreach.

Sustainability:

- **Market incentives:** Enterprises recover costs through rising seed and input sales.
- **Public support:** Seed enterprises linked to subsidies, training, and extension schemes.
- **Finance pathways:** MFIs/PKSF expand credit access; CSR/donor funds de-risk early investment.
- **Institutional uptake:** QA standards and delivery models adopted by DoF with budget alignment.

Milestones: 2026, priority districts operational; 2027, $\geq 80\%$ partners meet QA; 2028, nodes break even and DoF adopts QA for national roll-out.

2.2 Partnerships & Leverage

CGIAR champions. WorldFish (lead) develops improved carp, tilapia and SIS genetics and mass seed production systems under SAAF, while under S4I it provides technical leadership on decentralized delivery systems, MELIA, and nutrition integration. IFPRI will contribute policy research and advocacy to support embedding decentralized extension and QA models into national frameworks.

External partners. DoF is the institutional champion for certification, inspection, extension and compliance monitoring, with strong interest in embedding decentralized delivery within national procedures. BFRI, in collaboration with WorldFish and other stakeholders, develops standards for broodstock, performance, and biosecurity. Hatcheries, nurseries and seed traders organized under SeedCompacts supply verified seed to farmers, with several piloting mobile hatchery models in coastal zones. Input dealers and retailers extend the decentralized system by linking farmers to inputs and advisory services. Feed manufacturers such as Spectra Hexa Feeds Ltd are engaged to improve access to quality, affordable feed, complementing the genetics and seed delivery system. S4I engages NGOs by training their CHWs and equipping them to integrate fish-specific nutrition advice into ongoing community programs. The private digital sector supports OneFish platform, enabling farm-specific advisory, traceability, and market linkages. PKSf represents an emerging opportunity to expand outreach through microfinance partners and to pilot credit lines for hatcheries and nurseries. Processors and market actors such as Bengal Fish and Apon Welbeing strengthen demand by procuring fish from farmers and connecting them to consumer markets, reinforcing incentives for adoption.

Evidence of demand and commitment. DoF and BFRI are actively engaged in S4I activities, and WorldFish's role in the fisheries policy revision provides an entry point to institutionalize decentralized delivery models with supporting evidence. MoAs with private hatcheries ([Annex 1](#)) and DoF/BFRI ([Annex 2](#)) confirm commitment to dissemination of improved genetics with WorldFish. Private hatcheries are piloting mobile hatchery models; NGOs have nominated CHWs for training (Halder and Biswas, 2024); and discussions with PKSf are underway to explore financing partnerships.

Leverage. Continuation of WorldFish's core Genetics Program ensures ongoing improvement of carp, tilapia, and SIS lines. Bilateral initiatives provide strong alignment: the Asia–Africa Bluetech Superhighway (AABS) links with existing beneficiaries and private partners, while Sustho Sagor, led by WCS with WorldFish as a partner, complements outreach in coastal areas. Previous CSR financing (Standard Chartered, 2024) demonstrated potential for private finance, with opportunities for future CSR and blended finance engagement. Donor support and S4I catalytic funds (~USD 500k/year) will de-risk early scale-up by strengthening private-sector networks, training hatcheries and nurseries, enabling digital integration (OneFish app), and generating operational evidence for policy dialogue. These funds will consolidate lessons from AMD pilots, CSR initiatives, and early SeedCompacts into scalable models, ensuring evidence is institutionalized and leveraged by government, bilateral, and private-sector partners.

2.3 Risks & Bottlenecks

- **Governance risks.** Weak coordination among hatcheries, nurseries, and traders may undermine delivery quality. **Mitigation:** SeedCompacts will be formalized with clear roles, bylaws, certification, and digital traceability to maintain trust and seed authenticity.
- **Financial risks.** Limited capital for hatchery upgrades and nursery expansion may slow adoption. **Mitigation:** PKSf and MFIs will be engaged to expand credit for working capital, while CSR and donor co-finance represent opportunities to support early investment. S4I catalytic funds will de-risk scale-up through training, QA, digital integration, and MELIA, rather than provide direct working capital.
- **Policy risks.** Delays in DoF recognition or institutionalization of decentralized systems could hinder scale. **Mitigation:** Delivery models will be co-designed with DoF, supported by policy briefs and evidence through IFPRI advocacy, and aligned with the national fisheries policy revision.
- **Climate risks.** Floods, droughts, and salinity may disrupt seed supply chains. **Mitigation:** Mobile SIS hatcheries will be deployed, production sites diversified, and climate-resilient broodstock management adopted.
- **Equity and inclusion risks.** Women, youth, and marginalized households may be excluded. **Mitigation:** Gender and youth targets will be set; CHW-led nutrition messaging integrated; inclusion prioritized in climate-vulnerable areas; and partnerships with humanitarian actors (e.g., WFP) explored to adapt models for displaced settings.
- **Market risks.** Farmers may resist improved genetics due to lack of awareness or limited extension. **Mitigation:** Demonstrations will validate performance, CHWs and digital tools will drive demand creation, and traceability will assure quality.

Proactive risk management will ensure decentralized delivery systems remain inclusive, resilient, and credible, paving the way for sustainable institutionalization.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

3.1 Readiness & Responsible Scaling

Improved carp, tilapia and SIS genetics validated under SAAF are available for scale. This Solution Track builds on decentralized delivery systems that connect seed and input actors, digital platforms, and nutrition outreach to reach farmers reliably. These systems are rated at **Scaling Readiness Level 9** in CGIAR's PRMS, confirming maturity. Within this, SeedCompacts have demonstrated feasibility in Bangladesh pilots as a structured approach for organizing hatcheries, nurseries, and traders, ensuring verified seed delivery and farmer trust. CSR-backed pilots also confirmed private-sector willingness to co-invest in nursery and dealer upgrading, while OneFish has been tested for advisory and traceability. Together, these experiences establish readiness, with institutionalization by government as the next step.

Responsible scaling will be ensured through:

- **Inclusion:** $\geq 30\%$ women and youth in farming and service roles.
- **Nutrition:** SIS integration with CHWs delivering fish-based messages.
- **Environment:** Mobile hatcheries reduce groundwater use; carp–SIS integration supports ecosystem resilience.

3.2 Achievable Target, Resources & Activities

Achievable targets (2026–2028).

With S4I support and partner engagement, the track will:

- Strengthen 80 hatcheries and 4,800 nurseries within SeedCompacts, alongside 100 seed traders and 200 input dealers in the wider decentralized delivery system.
- Reach 546,000 farmers ($\geq 30\%$ women and youth) with verified seed, inputs, and advisory services.
- Expand access to nutrient-rich SIS seed in coastal areas through mobile hatcheries.
- Generate rigorous evidence with government on delivery performance, inclusivity, and cost-effectiveness.

Key resources.

- **Human:** S4I-funded trainers, MELIA specialists, and partner coaches to support delivery actors.
- **Institutional:** DoF oversight for extension; WorldFish development of QA protocols for SeedCompacts; DoF, BFRI and WorldFish standards and validation.
- **Financial:** ~USD 500k/year in S4I catalytic funds for training, digital tools, MELIA, and policy dialogue. WorldFish bilateral projects (e.g., AABS) will complement through joint meetings and scaling with existing beneficiaries. PKSf/MFI credit lines are under discussion, and CSR/donor co-financing is being explored.
- **Technical:** Rollout of the OneFish app, digital dashboards for seed traceability and adoption, MELIA protocols, and nutrition training materials.
- **Partnerships:** Collaboration with DoF, BFRI, PKSf, private hatcheries, ARITS as digital partner, NGOs with CHW programs, and bilateral initiatives such as AABS and WCS's Sustho Sagor.

Core activities. Capacity building for hatchery/nursery operators, traders, input dealers, and LSPs; establishment of SeedCompact bylaws, certification systems, and traceability tools; CHW training for nutrition messaging; MELIA at baseline (2026), midline (2028), and continuous feedback loops; and policy engagement with DoF and IFPRI to prepare for institutionalization.

3.3 Timeline, Milestones & Monitoring

2026. Baseline MELIA completed. Twenty hatcheries, 1,200 nurseries, and 25 seed traders organized into pilot SeedCompacts. DoF engagement initiated; BFRI, in collaboration with WorldFish and other stakeholders, providing standards. OneFish app piloted for farm-specific extension and traceability. CHWs from NGO nutrition programs trained with fish-specific

materials. Around **130,000 farmers** reached, covering **~60,000 ha** under improved aquaculture practices.

2027. Expansion to 20 more hatcheries, 1,200 nurseries, 25 seed traders, and 100 input dealers integrated into the wider decentralized extension system. Digital traceability tools tested with seed actors. Evidence on adoption and inclusivity presented to DoF. IFPRI-led advocacy supports integration of decentralized delivery networks into government programs, including subsidy and extension arrangements. Another **130,000 farmers** engaged, adding **~60,000 ha** under good practices.

2028. Midline MELIA conducted. Forty hatcheries, 2,400 nurseries, and 50 seed traders engaged in SeedCompacts, with 100 input dealers active in the wider system. Mobile SIS hatcheries operational in coastal areas. Cumulative **546,000 farmers** reached ($\geq 30\%$ women and youth), covering **~240,000 ha** with good aquaculture practices. Institutionalization advanced through government adoption of QA standards and cost-sharing for delivery, complemented by private co-financing.

Monitoring & learning. MELIA integrated with government and partner reporting. Digital dashboards track seed flows, QA compliance, and gender/youth participation. Household surveys through CHWs monitor nutrition outcomes. Adaptive learning loops ensure real-time adjustments, feeding evidence into delivery practices and DoF policy processes.

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Annexes

Annex 1: MoAs with 12 Private sector hatcheries

Annex 2: DoF/BFRI letter on G3 and G5 rohu trial and dissemination